SENTIMENT ANALYSIS FOR MENTAL HEALTH MONITORING

INTRODUCTION:

- Mental health disorders, especially depression, are becoming more common across all age groups.
- Unfortunately, many people hesitate to seek help due to cost, or lack of awareness.
- At the same time, millions of people share their thoughts, feelings, and struggles openly on social media.
- These posts often contain emotional cues that, if analyzed properly, can provide early signs of mental distress.
- This opens up a unique opportunity to use AI and Sentiment Analysis to monitor emotional health in a non-intrusive, scalable way.
- Our aim is to explore how text-based analysis can support mental health awareness and early detection.

What is Sentiment Analysis?

- Sentiment Analysis is a method in Artificial Intelligence that helps detect the emotional tone in written text.
- It uses Natural Language Processing (NLP) to determine whether a message is positive, negative, or neutral.
- Beyond just polarity, advanced models can also detect sadness, anger, joy, and fear.
- Example:
- “I feel so tired of everything” → Negative/Sad
- “Finally had a good day today” → Positive/Happy
- This technology allows us to analyze emotional patterns over time and detect changes that could indicate mental health struggles.
- It acts as the digital ear that listens to what people say online.

Goal of the study:

✦To develop a system that can:

- Identify negative emotional patterns in text data.
- Support early detection of mental health issues like depression or anxiety.
- Help mental health professionals by giving them data-driven insights.

✦This vision is not to diagnose, but to:

- Flag early signs.
- Encourage timely help.
- Raise awareness.
- Our approach is to use technology as a supportive tool in promoting mental well-being and reducing mental health risks.

Methodology:

Step 1: Data Collection

- Gather public, anonymized text data from online sources.
- Focus on personal expressions that could reflect mood or emotion.

Step 2: Preprocessing

- Clean the text: remove special characters, links, emojis, and stopwords.
- Normalize the data (lowercasing, lemmatization).

Step 3: Sentiment Analysis

- Detect emotions using AI models or pre-trained tools (e.g., VADER, BERT).

Step 4: Emotion Tracking

- Identify trends or shifts in emotional state across time.
- Highlight consistently negative expressions that may reflect depressive symptoms.

Step 5: Output

- Generate emotional scores for mental health indicators.

Computational Framework:

- Use labeled and anonymized mental health datasets from platforms like Twitter.
- Apply preprocessing steps such as tokenization, lemmatization, and stopword removal.
- Convert text into numerical vectors using TF-IDF, Word2Vec, or BERT.
- Train models including SVM, Logistic Regression, LSTM, GRU, and BERT.
- Classify the emotion as positive, negative, or depressive.
- Evaluate model performance using accuracy, precision, recall, and F1-score.
- Generate emotional scores and flag high-risk posts.

CONCLUSION:

- Sentiment analysis offers a modern, scalable approach to mental health monitoring.
- It helps detect emotional distress in real-time through language patterns.
- Can support wellness programs, therapy platforms, or early-warning systems.
- Not a replacement for therapy, but a powerful complement for proactive care.
- Enables early intervention by identifying users at emotional risk.
- Encourages a data-driven approach to understanding mental well-being.

Benefits:

- Scalable, real-time monitoring.
- Reaches people who may not seek help directly.

THANK YOU!